

Dehydroepiandrosterone (DHEA) Sulfate

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Work up women with infertility, amenorrhea, or hirsutism to identify the source of excessive androgen; aid in the evaluation of androgen excess (hirsutism and/or virilization), including Stein-Leventhal syndrome and adrenocortical diseases, including congenital adrenal hyperplasia and adrenal tumor. DHEA-S is not increased with hypopituitarism. It is low in Addison disease.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.³ In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.³ The test contains additives that minimize these effects.

Custom Additional Information

DHEA-S is a steroid hormone which is produced from the precursor cholesterol in the zona reticularis and broad fascia of the adrenal cortex.¹ The determination of elevated DHEA-S values is an important aid in the diagnosis of hirsutism and virilism.^{2,4} In addition to a differential diagnosis of hirsutism and virilism, further indications for this parameter are all forms of androgenization, hyperprolactinemia, polycystic ovarian syndrome, and the exclusion of an androgen-producing tumor of the adrenal cortex.² DHEA-S exhibits only a weak androgenic activity but can be metabolized to more active androgens, such as androstenedione and testosterone, which can indirectly cause hirsutism and virilism.^{2,5}

From 7 years of age onwards, an increase in DHEA-S levels is observed which then gradually after the age of 30 begins to fall again.⁶ Only elevated DHEA-S concentrations are of clinical importance; other factors which can be responsible for DHEA-S excess production are genetic enzyme defects of the adrenal cortex (adrenogenital syndrome),⁷ hyperplasia of the adrenal cortex, as well as androgen-producing tumors.²

The rate of secretion of DHEA-S into the blood stream is only slightly more than the rate observed for DHEA. As a consequence of the DHEA-S half-life of approximately one day, the DHEA-S level is, however, about a thousand-fold greater.⁸ DHEA-S is relatively strongly bound to albumin, only a small portion is nonprotein bound, and none appears to be bound to sex hormone-binding globulin (SHBG).⁹ Due to its high concentration and low inter- and intra-day variability, DHEA-S is an excellent indicator of adrenal cortex androgen production.^{8,10}

Together with testosterone, DHEA-S assays represent the assay of choice for initial screening tests to determine whether androgen values are elevated in hirsutism. Approximately 84% of the women suffering from hirsutism exhibit elevated androgen levels.¹¹ The main purpose of this is to exclude the presence of androgen-producing tumors (from the adrenal cortex or the ovaries).⁷

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.^{1,2}

Age	Male (µg/dL)	Female (µg/dL)
0 to 30 d	Not established	Not established
1 to 12 m	4.8–64.1	4.8–64.1
1 to 4 y	0.1–56.4	1.8–97.2
5 to 8 y	18.0–194.0	26.1–141.9
9 to 11 y	49.5–270.5	35.0–192.6
12 to 14 y	49.5–270.5	67.8–328.6
15 to 19 y	115.3–459.6	110.0–433.2
20 to 24 y	164.3–530.5	110.0–431.7
25 to 34 y	138.5–475.2	84.8–378.0
35 to 44 y	102.6–416.3	57.3–279.2
45 to 54 y	71.6–375.4	41.2–243.7
55 to 64 y	48.9–344.2	29.4–220.5
65 to 74 y	30.9–295.6	20.4–186.6
≥75 y	20.8–226.4	13.9–142.8

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

Footnotes

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